

Pooria Namyar

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EDUCATION

Ph.D. , University of Southern California, USA (advisor: Ramesh Govindan)	2025
M.Sc. , University of Southern California, USA	2023
B.Sc. , Sharif University of Technology, IRAN	2019

EXPERIENCE

Senior Researcher in AI Hardware & Infrastructure, Microsoft Research	Sep 2025 – Present
Research Intern , Google	Sep 2023 – Jan 2025
Research Intern , Microsoft Research	Jun 2022 – Jan 2023
Research Intern , Microsoft Research	Jun 2021 – Jan 2022

HONORS & AWARDS

ACM SIGCOMM Doctoral Dissertation Award	2026
Google Ph.D. Fellowship	2024 – 2025
Selected for NeTS Early Career Investigator Workshop , <i>National Science Foundation</i>	2025
Machine Learning and Systems Rising Star , <i>MLCommons</i>	2024
Ming Hsieh Institute (MHI) Scholarship , <i>University of Southern California</i>	2023 – 2024
Annenberg Graduate Student Fellowship , <i>University of Southern California</i>	2019 – 2023
Top 0.1% in Iranian Nation-wide University Entrance Exam for B.Sc.	2014

SELECTED RESEARCH

Efficient & Fault-Tolerant Pre-training 2025 – Present

- Lead a research effort on maximizing effective training time and reliability of large-scale LLM pretraining.
- Designed an asynchronous, per-iteration checkpointing system that reduces checkpoint transfer size by $3\times$ over the state of the art, adds $< 0.01\%$ step-time overhead, and bounds lost work to a single iteration.
- Built a near-zero-overhead observability framework using passive telemetry to detect and localize training performance degradation, including stragglers, in under one minute.
- Developing a near-zero-downtime failure recovery method using multi-tier hot standby GPUs ($< 1\%$ of fleet) and topology-aware, precomputed failover plans that reduce detection-to-first-healthy-step from a few minutes to under 10s.
- Developing training methods that inherently tolerate up to K stragglers with minor performance degradation.

Efficient Agentic Systems ([deployed at Microsoft](#)) 2025 – Present

- Developed cost-efficient and scalable model routers for agentic tool calling and coding agents, with production deployment at Microsoft. It processed trillions of customer tokens in three months and reduced costs by $\sim 75\%$.
- Developing a framework for synthesizing compact, cost-efficient, and task-specific agentic workflows.

Algorithmic Discovery & Robustness ([deployed at Microsoft](#)) 2022 – 2025

- Co-led a multi-year research effort on automated algorithm design, combining optimization, reinforcement learning, formal methods, and LLMs to discover, evaluate, explain, and improve algorithms.
- Developed adversarial discovery methods that automatically surface the worst-case behavior of algorithms and provide formal optimality guarantees.
- Integrated LLMs into the discovery loop to synthesize candidate algorithms, diagnose failure modes, and iteratively improve performance with formal bounds.
- Published eight top-tier papers, introduced tighter approximation ratios for long-standing theoretical problems, and produced several technology transfers to Microsoft Azure.

Robust & Scalable Resource Allocation Systems ([deployed at Microsoft](#); [adopted by Amazon](#)) 2019 – Present

- Developing market-based mechanisms to allocate and price Azure capacity for requests exceeding 100K cores, enabling Azure CVPs to systematically handle multi-billion-dollar requests.
- Developed a general and scalable max-min fair resource allocation method with formal guarantees; deployed in Microsoft's global network since 2023. Sped up the traffic engineering by 3x on average and up to 5.4x.
- Developed analytical methods to identify failures that can cause catastrophic impact on the WAN; deployed in Microsoft WAN capacity planning.
- Developed scalable techniques for accurately estimating effective datacenter capacity; Amazon applied these methods to [guide the design of its next-generation datacenters](#).
- Published seven papers in top-tier systems and networking venues and received three major research awards.

Clock Synchronization ([deployed at Google](#)) 2023 – 2025

- Developed a mathematical framework to analyze key properties of clock synchronization in data centers. Proved that running distributed clock consensus on a random overlay graph is scalable, converges quickly, and has a near-optimal error.
- Our work, Firefly, combines these analyses with system-level optimizations to achieve ≤ 10 ns error, and is deployed in [Google's trading system](#).

PUBLICATIONS

Under Prep.	Towards Near-Zero Downtime Failure Recovery for LLM Training Prasoon Sinha, Neeraja Yadwakhar, Kevin Hsieh, Pooria Namyar
Under Prep.	TrainSight: Inferring Training Characteristics from Passive Telemetry Pooria Namyar
Preprint	Switchcraft: AI Model Router for Agentic Tool Calling (link) Sharad Agarwal, Pooria Namyar , Alec Wolman, Rahul Ambavat, Ankur Gupta, Qizheng Zhang deployed at Microsoft
Preprint	Explanation-Guided Heuristic Design with LLMs (link) Pantea Karimi, Dany Rouhana, Pooria Namyar , Siva Kesava Reddy Kakarla, Venkat Arun, Behnaz Arzan
SIGCOMM 2026	Near-optimal Online Traffic Engineering Arvin Ghavidel, Pooria Namyar , Nikolai Matni, Walter Willinger, Ramesh Govindan
NSDI 2026	Heuristic Analysis from Source Code via Symbolic-Guided Optimization (link) Pantea Karimi, Siva Kakarla, Santiago Segarra, Ryan Beckett, Pooria Namyar , Mohammad Alizadeh, Behnaz Arzani
SIGCOMM 2025	Firefly: Scalable, Ultra-Accurate Clock Synchronization for Datacenters (link) Pooria Namyar , Yuliang Li, Weitao Wang, Nandita Dukkupati, KK Yap, Junzhi Gong, Chen Chen, Peixuan Gao, Devdeep Ray, Gautam Kumar, Yidan Ma, Ramesh Govindan, Amin Vahdat deployed at Google
SIGCOMM 2025	ZENITH: Towards A Formally Verified Highly-Available Control Plane (link) Pooria Namyar , Arvin Ghavidel, Mingyang Zhang, Harsha V. Madhyastha, Srivatsan Ravi, Chao Wang, Ramesh Govindan
SIGCOMM 2025	Raha: A General Tool to Analyze WAN Degradation (link) Behnaz Arzani, Sina Taheri, Pooria Namyar , Ryan Beckett, Siva Kakarla, Elnaz Jalilipour deployed at Microsoft
NSDI 2025	Enhancing Network Failure Mitigation with Performance-Aware Ranking (link) Pooria Namyar , Arvin Ghavidel, Daniel Crankshaw, Daniel Berger, Kevin Hsieh, Srikanth Kandula, Ramesh Govindan, Behnaz Arzani

NSDI 2025	Everything Matters in Programmable Packet Scheduling (link) Albert Gran Alcoz, Balázs Vass, <u>Pooria Namyar</u> , Behnaz Arzani, Gábor Rétvári, Laurent Vanbever
NSDI 2024	Solving Max-Min Fair Resource Allocations Quickly on Large Graphs (link) <u>Pooria Namyar</u> , Behnaz Arzani, Srikanth Kandula, Santiago Segarra, Daniel Crankshaw, Umesh Krishnaswamy, Ramesh Govindan, Himanshu Raj deployed at Microsoft
NSDI 2024	Finding Adversarial Inputs for Heuristics using Multi-level Optimization (link) <u>Pooria Namyar</u> , Behnaz Arzani, Ryan Beckett, Santiago Segarra, Himanshu Raj, Umesh Krishnaswamy, Ramesh Govindan, Srikanth Kandula led to improvement or deprecation of production heuristics at Microsoft
HotNets 2022	End-to-End Performance Analysis of Learning-enabled Systems (link) <u>Pooria Namyar</u> , Michael Schapira, Ramesh Govindan, Santiago Segarra, Ryan Beckett, Siva Kakarla, Behnaz Arzani
HotNets 2024	Towards Safer Heuristics With Xplain (link) Pantea Karimi, Solal Pirelli, Siva Kakarla, Ryan Beckett, Santiago Segarra, Beibin Li, <u>Pooria Namyar</u> , Behnaz Arzani
HotNets 2022	Minding the gap between Fast Heuristics and their Optimal Counterparts (link) <u>Pooria Namyar</u> , Behnaz Arzani, Ryan Beckett, Santiago Segarra, Himanshu Raj, Srikanth Kandula
INFOCOM 2022	Optimal Oblivious Routing for Structured Networks (link) Sucha Supittayapornpong, <u>Pooria Namyar</u> , Mingyang Zhang, Minlan Yu, Ramesh Govindan
SIGCOMM 2021	A Throughput-Centric View of the Performance of Datacenter Topologies (link) <u>Pooria Namyar</u> , Sucha Supittayapornpong, Mingyang Zhang, Minlan Yu, Ramesh Govindan adopted by Amazon

PATENTS

1. Flow Network Intermediate Representation For Optimization Problems, US patent application, Published 6/2025
2. Solving Max-min Fair Resource Allocation at Large Scale, US patent application, Published 9/2024
3. Network Traffic Control Using Estimated Maximum Gap, US patent application, Published 3/2024
4. Impact-Aware Mitigation For Computer Networks, US patent application, Published 3/2023
5. Artificial Intelligence Analysis For Holistic System Design, US patent application, under review
6. Network Performance Degradation Analyses And Remediation, US patent application, under review
7. Explanation-guided Heuristic Design, US patent application, under review

INVITED TALKS & PRESENTATIONS

Toward Reliable and Scalable Network Algorithms

Microsoft Research, Redmond, WA, USA	Mar 2025
University of Maryland, College Park, MD, USA	Mar 2025

Heuristic Analyzers for Networked Systems

ACM HotNets 2024, Irvine, CA, USA (Ph.D. short talk)	Nov 2024
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End-to-End Performance Analysis of Learning-enabled Systems

ACM HotNets 2024, Irvine, CA, USA	Nov 2024
Microsoft Research, Virtual	Nov 2024

Finding Adversarial Inputs for Heuristics using Multi-level Optimization

Rising Stars workshop at NVIDIA, CA, USA	Jul 2024
Google Systems Talk, Sunnyvale, CA, USA (Host: Jeff Mogul)	Jun 2024
NSDI 2024, Santa Clara, CA, USA	Apr 2024
Microsoft Research, Virtual	Apr 2024

Solving Max-Min Fair Resource Allocation at Large Scale

NSDI 2024, Santa Clara, CA, USA

Apr 2024

Microsoft Research, Virtual

Sep 2022

Minding the gap between Fast Heuristics and their Optimal Counterparts

ACM HotNets 2022, Austin, Texas, USA

Nov 2022

Microsoft Research, Virtual

Sep 2022

A Throughput-Centric View of the Performance of Datacenter Topologies

Google Networking Research Summit, Virtual

Mar 2022

11th Annual Research Festival, University of Southern California, USA

Oct 2021

Microsoft Research, Virtual

Aug 2021

ACM SIGCOMM 2021 Conference, Virtual

Aug 2021

TEACHING

Teaching

- Teaching Assistant, Introduction to Internetworking (CSCI 353), Fall 2024, University of Southern California
- Guest Lecturer, Networked Systems in Cloud Computing (CS 599/656), Spring 2024, University of Southern California
- Teaching Assistant, Networked Systems in Cloud Computing (CS 599/656), Spring 2024, University of Southern California
- Teaching Assistant, Data Networks, Fall 2018, Sharif University of Technology
- Teaching Assistant, Introduction to Machine Learning, Fall 2018, Sharif University of Technology
- Teaching Assistant, EE Principles Laboratory, Fall 2016, Sharif University of Technology

Mentoring

- Prasoon Sinha (Ph.D. at University of Texas Austin), Summer 2026 – Present
- Adam Lechowicz (Ph.D. at University of Massachusetts Amherst), Summer 2026 – Present
- Arvin Ghavidel (Ph.D. at University of Southern California), Fall 2022 – Present
- Mahdi Alizadeh (Ph.D. at University of Southern California), Spring 2026 – Present
- Roozbeh Bostandoost (Ph.D. at University of Massachusetts Amherst), Spring 2024 – Present
- Pantea Karimi (Ph.D. at Massachusetts Institute of Technology), Fall 2024 – Present
- Neal Sharma (BS from University of California, Berkeley, 2022) → M.S. student at University of California, Irvine
- Smit Kadvani (M.S. from University of Southern California, 2021) → Machine Learning Engineer at Meta

PROFESSIONAL SERVICE

Service

- Program Committee: ACM SIGCOMM 2026, ATC 2026, CoNEXT 2026, HotNets 2026, APNet 2026
- Pre-Review Task Force: NSDI 2025
- AE Reviewer: ACM SIGCOMM 2024, ACM SOSP 2023, ACM SIGCOMM 2023
- Reviewer: IEEE SPAWC 2024, ACM TOCS 2026
- External Reviewer: NSDI 2022, ESA 2025

Outreach

- Viterbi Graduate Mentor, University of Southern California, 2024 – 2025
- Volunteer in [CS@SC Coding Camps](#), Viterbi K-12 STEM Center, 2024
- Panelist, Sharing PhD journey with Undergraduate Students, University of Southern California, 2024
- Womxn in Engineering (WIE) mentor, University of Southern California, 2023 – 2024
- Organizer, MHI Annual Research Festival, 2023

MEDIA COVERAGE

- Urs Hölzle's post (Oct 23, 2025) about [our work on clock synchronization](#).
- Viterbi School of Engineering (Oct 3, 2024), [Ph.D. Candidate Pooria Namyar Among 41 Named MLCommons Rising Stars](#).
- Viterbi School of Engineering (Jul 10, 2024), [Student Builds an 'Angel' to Lift Up Microsoft's Wide-Area Network](#).
- Microsoft Research Blog (Jun 20, 2024), [Microsoft's official announcement about the deployment of Soroush](#).
- Microsoft Research Blog (Jan 23, 2024), [MetaOpt: Examining, explaining, and improving heuristic performance](#).